

ShopSphere E-Commerce API Documentation

Complete API Reference for Manual Testing in Postman and Swagger

Base URL: `http://localhost:8080`

Controllers: Auth, Product, Category, Wishlist, Cart, and Order.

1. Authentication & Authorization

Register and login are public. Cart, Wishlist, and Order endpoints require authentication. Product POST, PUT, and DELETE require ADMIN access.

JWT Token

For protected APIs, login first and copy the JWT token. In Postman choose Authorization → Bearer Token and paste the token.

```
Authorization: Bearer <your-jwt-token>
```

2. Auth Controller — /auth

POST /auth/register

URL: `http://localhost:8080/auth/register`

Public endpoint

```
{
  "name": "Deepika",
  "email": "deepika@example.com",
  "password": "Password@123"
}
```

Expected: 201 Created

POST /auth/login

URL: `http://localhost:8080/auth/login`

Public endpoint

```
{
  "email": "deepika@example.com",
  "password": "Password@123"
}
```

Expected: 200 OK with login response containing JWT token.

3. Category Controller — /categories

POST /categories

URL: `http://localhost:8080/categories`

```
{
  "name": "Electronics"
}
```

GET /categories

URL: `http://localhost:8080/categories`

No request body.

4. Product Controller — /products

GET endpoints retrieve products. POST, PUT, and DELETE require an ADMIN Bearer token.

Create Product

POST /products

URL: `http://localhost:8080/products`

```
{
  "name": "Wireless Headphones",
  "description": "Bluetooth over-ear headphones",
  "price": 2999.00,
  "stockQuantity": 25,
  "available": true,
  "brand": "Sony",
  "categoryId": 1
}
```

Product URLs

Endpoint	Example URL
GET /products	http://localhost:8080/products?page=0&size;=10&sort;=id,asc
GET /products/{id}	http://localhost:8080/products/1
PUT /products/{id}	http://localhost:8080/products/1
DELETE /products/{id}	http://localhost:8080/products/1
GET /products/brand/{brand}	http://localhost:8080/products/brand/Sony
GET /products/price/less-than/{price}	http://localhost:8080/products/price/less-than/5000
GET /products/price/greater-than/{price}	http://localhost:8080/products/price/greater-than/1000
GET /products/price/less-than-equal/{price}	http://localhost:8080/products/price/less-than-equal/5000
GET /products/price/greater-than-equal/{price}	http://localhost:8080/products/price/greater-than-equal/1000
GET /products/available	http://localhost:8080/products/available
GET /products/category/{categoryName}	http://localhost:8080/products/category/Electronics
GET /products/search?brand=Sony&available;=true	http://localhost:8080/products/search?brand=Sony&available;=true
GET /products/search/name?keyword=phone	http://localhost:8080/products/search/name?keyword=phone
GET /products/category-price?...	http://localhost:8080/products/category-price?categoryName=Electronics&price;=5000
GET /products/category-brand?...	http://localhost:8080/products/category-brand?categoryName=Electronics&brand;=Sony
GET /products/category-brand/available?...	http://localhost:8080/products/category-brand/available?categoryName=Electronics&brand;=Sony
GET /products/category-price/available?...	http://localhost:8080/products/category-price/available?categoryName=Electronics&price;=5000
GET /products/category-brand-price?...	http://localhost:8080/products/category-brand-price?categoryName=Electronics&brand;=Sony&price;=5000
GET /products/filter?...	http://localhost:8080/products/filter?categoryName=Electronics&brand;=Sony&price;=5000

5. Cart Controller — /cart

All endpoints require Bearer authentication.

Endpoint	Example URL	Purpose
POST /cart/{productId}?quantity={quantity}	http://localhost:8080/cart/1?quantity=2	Add product
GET /cart	http://localhost:8080/cart	Get current cart
PUT /cart/{productId}?quantity={quantity}	http://localhost:8080/cart/1?quantity=5	Update quantity
DELETE /cart/{productId}	http://localhost:8080/cart/1	Remove product
DELETE /cart	http://localhost:8080/cart	Clear cart

Important: Use /cart/1?quantity=2, not /cart/add. The controller expects productId as a Long path variable.

6. Wishlist Controller — /wishlist

All endpoints require Bearer authentication.

Endpoint	Example URL	Purpose
POST /wishlist/{productId}	http://localhost:8080/wishlist/1	Add product
GET /wishlist	http://localhost:8080/wishlist	Get wishlist
DELETE /wishlist/{productId}	http://localhost:8080/wishlist/1	Remove product

7. Order Controller — /orders

All endpoints require Bearer authentication.

POST /orders

URL: http://localhost:8080/orders

Body: None. Places an order using the authenticated user's current cart.

Before placing an order, the cart must contain items, products must be available, and stock must be sufficient.

GET /orders

URL: http://localhost:8080/orders

Body: None. Returns orders belonging to the authenticated user.

8. Recommended Postman Testing Flow

1. Register a user.
2. Login and copy the JWT token.
3. Set the JWT as Bearer Token for protected APIs.
4. Create a category.
5. Create product(s) with an ADMIN user if product security is enabled.
6. Verify product CRUD and product search APIs.
7. Add products to the cart.
8. Verify the cart.
9. Test wishlist operations.
10. Place an order.

11. Verify orders using GET /orders.

9. Common Status Codes

Status	Meaning
200 OK	Request successful
201 Created	Resource created
204 No Content	Delete/clear successful
400 Bad Request	Invalid data, validation, or business rule failure
401 Unauthorized	Token missing, invalid, or expired
403 Forbidden	Authenticated but insufficient permission
404 Not Found	Resource not found

10. Swagger

Swagger UI: <http://localhost:8080/swagger-ui/index.html>

OpenAPI JSON: <http://localhost:8080/v3/api-docs>